The Double-Ended Helix: A Three-Dimensional Geometric Tool for the Vanishing Points of Dirichlet L-Functions

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Abstract

We present a three-dimensional geometric tool: a double-ended chiral helix carrier on which a phasor fiber identifies the vanishing points — the crossings where the phasor accumulation cancels — that coincide with the zeros of the Riemann zeta function and Dirichlet L-functions. The carrier is a growing-radius helix with integers placed at uniform arclength spacing $\Delta = \pi/3$; a fiber rides it and accumulates phasors. The fiber attached to a Dirichlet character χ is the partial-sum sequence $\sum_{n<N} \chi(n)\, n^{-s}$; its limit is the Dirichlet L-function $L(s, \chi)$ on $\operatorname{Re} s > 1$, and, after Abel summation, on the whole half-plane $\operatorname{Re} s > 0$, the critical line included. The angular winding of the carrier is a completely multiplicative character built from prime angles by the fundamental theorem of arithmetic, with no logarithm; the logarithm enters only in the external bridge $\operatorname{wind} n \leftrightarrow n^{it}$ that identifies the geometric winding with the analytic phasor. We record the geometric explicit formula (the logarithmic derivative of the fiber is the winding-weighted von Mangoldt prime field, and each simple zero is a simple pole with an explicit residue), the reality of the completed object on the critical line (read through a projection that rescales the helix's $\pi/3$ gauge to a unit coordinate), the conjugacy of the two chiralities of the double-ended carrier, and a determinant identity: at a conjugate crossing the two chiral eigenphases $z = e^{-iy \log n}$ and $\overline{z}$ assemble the transverse block $\operatorname{diag}(z, \overline{z})$, whose determinant is $z\,\overline{z} = |z|^2 = 1$. These two chiral eigenphases are the values of a unit-modulus eigenstate of the self-adjoint generator $D = -i\, d/dt$ (real eigenvalue, by von Neumann reality), and under the de Branges change of variables the on-line cancellations we find are real spectral points of a Hermite–Biehler structure function. We do not claim that the structure function for Λ is Hermite–Biehler — equivalently, that there are no off-line zeros — which would be the Riemann Hypothesis and is left open; the vanishing $\Rightarrow$ eigenstate link is likewise taken as a hypothesis. The coincidence of the fiber's vanishing points with the actual zeros has been verified numerically for the leading zeros of ζ and ten Dirichlet L-functions. Every statement below is formalized in Lean 4 with Mathlib and checked by the kernel; the axiom footprint of each named theorem is {propext, Classical.choice, Quot.sound}, with no `sorry` and no additional axioms.

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1 Objects and notation

Let χ be a Dirichlet character modulo q and $L(s,\chi)$ the associated L-function; ζ is the Riemann zeta function and $\Lambda(s) = \texttt{completedRiemannZeta}(s)$ its completed form, normalized so that $\Lambda(1-s) = \Lambda(s)$. The *critical line* is the set $s = \frac{1}{2} + iy$, $y \in \mathbb{R}$. We write $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ for the unit circle (Mathlib `Circle`) and $\overline{\cdot}$ for complex conjugation ($\texttt{starRingEnd}\,\mathbb{C}$). For an arithmetic weight χ the *phasor* attached to the integer $n \geq 1$ on the vertical line $s = \sigma + iy$ is $\chi(n)\, n^{-s}$.

The classical analytic theory of ζ and Dirichlet L-functions — the Euler product, the functional equation, and the von Mangoldt explicit formula [1, 2, 3, 4, 5, 7] — enters here through the corresponding Mathlib lemmas.

All named results carry the identifier of the corresponding Lean declaration, e.g. `frobenius_conjugate_det_one`; Section 12 records the verification.

2 The $\pi/3$ helix carrier

2.1 The space curve and its arclength

Definition 2.1 (Carrier curve; `Geometry.helix`, `helixVel`, `speed`). *For pitch p and radial rate r, the carrier is the growing-radius helix*

$$\gamma(k) \;=\; \big(r\,k\cos 2\pi k,\; r\,k\sin 2\pi k,\; p\,k\big), \qquad k \in \mathbb{R}.$$

Its velocity is $\gamma'(k) = \big(r\cos 2\pi k - 2\pi r k \sin 2\pi k,\ r\sin 2\pi k + 2\pi r k\cos 2\pi k,\ p\big)$*, and the squared speed is*

$$\|\gamma'(k)\|^2 \;=\; p^2 + r^2 + (2\pi r k)^2.$$

The climber *is* $k(y) = e^y/p$*, so the height is* $z = p\,k(y) = e^y$ *and the cylindrical radius is* $R(y) = r\,k(y) = (r/p)e^y$*.*

Definition 2.2 (Arclength; `Geometry.arclength`, `arclengthClosed`)**.** *The arclength of* γ *from* 0 *to* k *is* $S(k;p,r) = \int_0^k \sqrt{p^2 + r^2 + (2\pi r t)^2}\,dt$*.*

Theorem 2.3 (Closed-form arclength; `arclength_closed_form`, `arclength_r_zero`)**.** *For* $r > 0$*,*

$$S(k;p,r) \;=\; \frac{k}{2}\sqrt{p^2 + r^2 + 4\pi^2 r^2 k^2} \;+\; \frac{p^2 + r^2}{4\pi r}\ \operatorname{arsinh}\Big(\frac{2\pi r k}{\sqrt{p^2 + r^2}}\Big),$$

and $S(k;p,0) = p\,k$ *for* $p \geq 0$*.*

Proof sketch. Differentiate the stated antiderivative and check it equals the integrand $\sqrt{p^2 + r^2 + 4\pi^2 r^2 k^2}$ pointwise, then apply the fundamental theorem of calculus on $[0,k]$; the constant-pitch case is $\int_0^k p\,dt$. $\square$

2.2 Uniform integer placement and the area law

Definition 2.4 (Spacing, index, integer angle; `Geometry.Delta`, `Nindex`, `spinAngle`)**.** *The fixed spacing is* $\Delta = \pi/3$*. The continuous geometric index is* $N(y) = S(k(y);p,r)/\Delta$*, so* $N(y) = (3/\pi)\,S(k(y);p,r)$ *(`Nindex_eq`). The integer angular coordinate is* $s_n = n\cdot(\pi/3)$*.*

Theorem 2.5 (Uniform $\pi/3$ spacing; `carrier_spacing`)**.** *For every* $n \in \mathbb{N}$*,* $s_{n+1} - s_n = \pi/3$*.*

Theorem 2.6 (Area-law bound; `arclengthClosed_lower_bound`, `arclengthClosed_upper_bound`)**.** *For* $r > 0$ *and* $k \geq 0$*,*

$$\pi r\,k^2 \;\leq\; S(k;p,r) \;\leq\; \pi r\,k^2 + 2k\sqrt{p^2 + r^2}.$$

Definition 2.7 (Wound integer placement; `Geometry.windParameter`, `windIntegerSite`, `exists_unique_windParameter`)**.** *Integer* n *sits at arclength* $s_n = n\Delta$ *on the unwound line (Definition 2.4).* Winding the line onto the helix *places it at the unique parameter* $k_n \geq 0$ *with* $S(k_n;p,r) = n\Delta$ *(`exists_unique_windParameter`; the map* $s \mapsto k_n$ *is `windParameter`). The wound site is* windIntegerSite$(n) = \gamma(k_n)$*, with cylindrical radius* $r_n = r\,k_n$ *(`windIntegerSite_cyl_radius`).*

Theorem 2.8 (Emergent radius / area law; `emergent_radius_sq_tendsto`, `windIntegerSite_radius_sq_tendsto`)**.** *For the wound placement* $S(k_n;p,r) = n\Delta$ *of Definition 2.7,*

$$\frac{r_n^2}{n} \;=\; \frac{(r\,k_n)^2}{n} \;\xrightarrow[n\to\infty]{}\; \frac{r\Delta}{\pi}, \qquad \textit{equivalently} \qquad r_n \;\sim\; \sqrt{\tfrac{r\Delta}{\pi}}\,\sqrt{n},$$

so the enclosed area $\pi r_n^2 \sim r\Delta\,n$ *is linear in* n*. The factor* $\sqrt{n}$ *is thus* derived from the arclength geometry*: the bound of Theorem 2.6 forces* $S(k) \propto k^2$*, hence* $k_n \propto \sqrt{n}$ *and* $r_n = r\,k_n \propto \sqrt{n}$ *— it is not posited.*

Remark 2.9 (Unit gauge; `windIntegerSite_radius_sq_tendsto_unit_gauge`). *The area-law constant is $r\Delta/\pi$; with the native spacing $\Delta = \pi/3$ it equals $r/3$, so $r_n \sim \sqrt{n}$* exactly *in the gauge $r\Delta = \pi$, i.e. $r = 3$ (`windIntegerSite_radius_sq_tendsto_unit_gauge`); the companion implementation's plot uses $r = 1$, giving the constant $1/\sqrt{3}$. In this unit gauge the carrier point* $\mathrm{helixPt}(n) = \sqrt{n}\cdot \mathrm{wind}(n)$ *(Definition 3.1) has modulus* $\|\mathrm{helixPt}(n)\| = \sqrt{n}$ *(`HelixLogFree.norm_helixPt`), matching the emergent radius r_n asymptotically (`helixPt_radius_matches_areaLaw`: the ratio* $\|\mathrm{helixPt}(n)\|/r_n \to 1$ *in this gauge); the winding contributes unit modulus (Theorem 3.2).*

2.3 The scale-critical exponent

The area law gives more than the radius $\sqrt{n}$: it pins the critical exponent. Write $r_n = \mathrm{carrierRadius}(n)$ for the wound radius of Definition 2.7 (`Geometry.carrierRadius`), and pair it with a candidate weight $n^{-\sigma}$, $\sigma \in \mathbb{R}$.

Theorem 2.10 (The critical exponent is scale-critical; `sigma_half_is_scale_critical`). *For $\sigma \in \mathbb{R}$, the scale-balance product $n^{-\sigma} r_n$ has a positive limit if and only if $\sigma = \frac{1}{2}$:*

$$\bigl(\exists L > 0 :\ n^{-\sigma} r_n \xrightarrow[n\to\infty]{} L\bigr) \qquad \Longleftrightarrow \qquad \sigma = \tfrac{1}{2}.$$

Concretely, $n^{-\sigma} r_n \to 0$ for $\sigma > \frac{1}{2}$ (the weight decays faster than the radius grows), $n^{-\sigma} r_n \to +\infty$ for $\sigma < \frac{1}{2}$ (the weight outruns the radius), and $n^{-\sigma} r_n \to \sqrt{r\Delta/\pi} > 0$ for $\sigma = \frac{1}{2}$. Thus $\frac{1}{2}$ is the unique exponent at which the weight reciprocates the emergent radius. It is gauge-independent*: the gauge $r\Delta$ sets only the value of the limit, never which σ is critical.*

Proof sketch. By Theorem 2.8, $r_n/\sqrt{n} \to \sqrt{r\Delta/\pi} =: c > 0$. Factor $n^{-\sigma} r_n = (r_n/\sqrt{n}) \cdot n^{1/2-\sigma}$; the first factor tends to c, and the second tends to 0, to 1, or to $+\infty$ according as $\sigma > \frac{1}{2}$, $\sigma = \frac{1}{2}$, or $\sigma < \frac{1}{2}$. Only $\sigma = \frac{1}{2}$ yields a positive finite limit; uniqueness of limits closes both directions. ☐

Remark 2.11. *This is the geometric origin of the critical line: the $\frac{1}{2}$ of* $\mathrm{Re}\, s = \frac{1}{2}$ *is the $\frac{1}{2}$ of $r_n \sim \sqrt{n} = n^{1/2}$. It locates the line on which the phasor weights are scale-normalized to $O(1)$, so that the accumulation is governed by the winding (the phase cancellation) rather than the amplitude — which is exactly the mechanism by which the vanishing points arise (Section 5). It is a statement about the carrier alone; that the zeros of ζ and L then* lie *on this line is the separate analytic content of the functional equation (Theorem 9.3), not of the area law.*

2.4 The mod-6 bucket structure

Because $6 \cdot (\pi/3) = 2\pi$, the integer angle s_n is 6-periodic on the circle.

Theorem 2.12 (Bucket periodicity and counts; `spin_phasor_period6`, `bucket_count`). $\exp(i\, s_{n+6}) = \exp(i\, s_n)$ *for all n, and for $a < 6$ exactly N of the integers in $[0, 6N)$ satisfy $n \equiv a$* (mod 6)*. Thus the integers collapse onto six angular directions, each receiving an equal share.*

Remark 2.13 (Sixth roots of unity; Eisenstein integers). *The integer-angle phasor* $\exp(i\, s_n) = \exp(i\, n\pi/3)$ *takes values in the sixth roots of unity $\mu_6 = \{e^{ik\pi/3} : 0 \le k < 6\}$, cycling through them as n runs mod 6; the primitive sixth root $\zeta = e^{i\pi/3}$ ($\zeta^2 = \zeta - 1$, the cyclotomic relation Φ_6) generates μ_6. The ring $\mathbb{Z}[\zeta]$ is the ring of Eisenstein integers [8] — a hexagonal (triangular) lattice whose unit group is exactly μ_6. So the $\pi/3$ gauge is the Eisenstein symmetry, and the six buckets of Theorem 2.12 are its units.*

3 The log-free winding and the bridge

3.1 The fundamental-theorem-of-arithmetic winding

Definition 3.1 (Winding angle and winding; `HelixLogFree.windAngle`, `wind`, `helixPt`). *Fix a prime-angle assignment θ (in Lean, a function $\theta : \mathbb{N} \to \mathbb{R}$; only its values on primes enter). The* winding angle *is the completely additive extension read off the prime factorization,*

$$\Theta(n) \;=\; \sum_{p^e \,\|\, n} e\,\theta(p) \;=\; \texttt{n.factorization.sum}\,(\lambda\, p\, e.\; e \cdot \theta(p)),$$

and the winding *is* $\mathrm{wind}(n) = \exp(i\,\Theta(n)) \in \mathbb{T}$. *The carrier point is* $\mathrm{helixPt}(n) = \sqrt{n} \cdot \mathrm{wind}(n)$. *No logarithm occurs in* Θ.

Theorem 3.2 (Multiplicative character; `windAngle_mul`, `wind_mul`, `wind_one`). *For $m, n \neq 0$,* $\Theta(mn) = \Theta(m) + \Theta(n)$ *and* $\mathrm{wind}(mn) = \mathrm{wind}(m)\,\mathrm{wind}(n)$, *with* $\mathrm{wind}(1) = 1$. *Thus* $\mathrm{wind} : \mathbb{N} \to \mathbb{T}$ *is a completely multiplicative character [6].*

Proof sketch. Θ is a sum over `Nat.factorization`; additivity is `Nat.factorization_mul` (the fundamental theorem of arithmetic). Exponentiating turns additivity into multiplicativity on $\mathbb{T}$ via `Circle.exp_add`. □

3.2 The bridge $\mathrm{wind}\, n \leftrightarrow n^{it}$

The single place a logarithm enters is the external identification of the geometric winding with the analytic phasor.

Theorem 3.3 (Logarithm is FTA-additive; `real_log_eq_factorization_sum`). *For $n \neq 0$, $\log n = \sum_{p^e \,\|\, n} e \,\log p$.*

Theorem 3.4 (The bridge; `windAngle_glog`, `wind_glog_eq_cpow`). *With the prime-angle assignment $\theta(p) = \gamma \log p$, one has $\Theta(n) = \gamma \log n$ and hence*

$$\mathrm{wind}(n) \;=\; n^{i\gamma} \qquad (n \geq 1).$$

Proof sketch. Substitute Theorem 3.3 into Θ to get $\Theta(n) = \gamma \log n$; then $\exp(i\gamma \log n) = n^{i\gamma}$ by the principal-branch definition of $n^{i\gamma}$ for the positive real base n. □

4 The phasor fiber and its accumulation

4.1 Phasor decomposition

Theorem 4.1 (Vertical-line phasor and magnitude; `cpow_vertical_line_phasor`, `norm_cpow_vertical_line`). *For a positive real base x and $s = \sigma + iy$,*

$$x^{-s} \;=\; x^{-\sigma}\, \exp\!\big(-(y \log x)\, i\big), \qquad \big|x^{-s}\big| \;=\; x^{-\sigma}.$$

In particular, on the critical line $n^{-(1/2+iy)} = n^{-1/2} \exp(-(y \log n) i)$ with magnitude $n^{-1/2}$ for every y (`cpow_critical_line`, `norm_cpow_critical_line`).

Definition 4.2 (The spin; `LFunctionPhasor.spin`). *The* spin *of the integer n at ordinate y is the unit-modulus factor* $\mathrm{spin}(y, n) = \exp(-(y \log n)\, i)$.

Theorem 4.3 (The fiber rides the carrier; `fiber_rides_carrier`). *Write* $\mathrm{wind}_{-t\log}$ *for the winding of Definition 3.1 with the assignment* $\theta(p) = -t\log p$, *so* $\mathrm{wind}_{-t\log}(n) = n^{-it}$ *by the bridge (Theorem 3.4). For* $n \geq 1$, *the carrier magnitude* $n^{-1/2}$ *— the reciprocal of the emergent radius* $\sqrt{n}$ *— times this winding is the Dirichlet phasor:*

$$n^{-1/2} \cdot \mathrm{wind}_{-t\log}(n) \;=\; n^{-(1/2+it)}.$$

Remark 4.4 (The fiber's exponent is scale-critical; `critical_exponent_is_scale_critical`). *The exponent* $\frac{1}{2}$ *here is not a chosen constant. By Theorem 2.10 — re-exported at the unit-gauge carrier* $r = 3$ *as* `critical_exponent_is_scale_critical` *— it is the unique* σ *for which the fiber amplitude* $n^{-\sigma}$ *scale-balances the area-law radius* r_n. *So the critical line* $\operatorname{Re} s = \frac{1}{2}$ *that the fiber rides is the scale-balance locus of the carrier, forced by the geometry rather than inserted.*

4.2 Accumulation to L

Definition 4.5 (Finite phasor carrier; `DirichletPhasorCarrier.finiteCarrier`, `phasorTerm`). $G_{\chi,N}(s) = \sum_{n<N} \chi(n)\, n^{-s}$, *the partial sums of the carrier-riding phasors.*

Theorem 4.6 (Accumulation on $\operatorname{Re} s > 1$; `fiber_accumulates_to_L`, `finiteCarrier_tendsto_LFunction`). *For every Dirichlet character* χ *and* $\operatorname{Re} s > 1$, $G_{\chi,N}(s) \to L(s,\chi)$ *as* $N \to \infty$.

5 Strip extension: accumulation onto the critical line

The accumulation converges beyond the region of absolute convergence, down to $\operatorname{Re} s > 0$.

Theorem 5.1 (Dirichlet strip extension; `dirichlet_strip_tendsto_LFunction`). *For a Dirichlet character* $\chi \neq 1$ *modulo* q *and* $\operatorname{Re} s > 0$,

$$\sum_{n<N} \chi(n)\, n^{-s} \;\xrightarrow[N\to\infty]{}\; L(s,\chi).$$

Theorem 5.2 (Eta strip extension; `eta_strip_tendsto`). *For* $\operatorname{Re} s > 0$ *and* $s \neq 1$,

$$\sum_{n<N} (-1)^{\,n+1}\, n^{-s} \;\xrightarrow[N\to\infty]{}\; \big(1 - 2^{\,1-s}\big)\zeta(s).$$

Proof sketch. Bucket cancellation gives a uniform bound on the character sums $\sum_{n<N} \chi(n)$; Abel summation by parts [6, 3] then yields convergence of $\sum \chi(n) n^{-s}$ on $\operatorname{Re} s > 0$, and the limit agrees with $L(s,\chi)$ on $\operatorname{Re} s > 1$ and hence everywhere on the connected strip by the identity theorem. For the alternating weight the factor $1 - 2^{1-s}$ is the even-term correction relating η and ζ. □

Theorem 5.3 (Continuous reach to the line; `continuous_model_zeta`). *For every* $y \in \mathbb{R}$, *at* $s = \frac{1}{2} + iy$,

$$\sum_{n<N} (-1)^{\,n+1}\, n^{-s} \;\longrightarrow\; \big(1 - 2^{\,1-s}\big)\zeta\big(\tfrac{1}{2} + iy\big),$$

and the limit is 0 *if and only if* $\zeta(\frac{1}{2} + iy) = 0$.

Proof sketch. Theorem 5.2 at $s = \frac{1}{2} + iy$ (using $\operatorname{Re} s = \frac{1}{2} > 0$ and $s \neq 1$). On the line the factor $1 - 2^{1-s}$ is nonzero (`correction_factor_ne_zero_on_critical_line`), so the product vanishes exactly when $\zeta(\frac{1}{2} + iy)$ does (`etaTrivial_eq_zero_iff`). □

6 The geometric explicit formula

The logarithmic derivative of the fiber is the prime-power accumulation weighted by the winding.

Theorem 6.1 (Prime field; `explicit_formula_prime_field`, `dirichlet_logDeriv_eq_tsum`). *For* $\operatorname{Re} s > 1$*,*

$$-\frac{L'(s,\chi)}{L(s,\chi)} \;=\; \sum_n \chi(n)\,\Lambda(n)\,n^{-s},$$

where Λ *is the von Mangoldt function. The zeros of* $L(\cdot,\chi)$ *are the poles of the meromorphic continuation of this object.*

Definition 6.2 (Residue harmonic; `Residue.residueHarmonic`). *For* $x > 0$ *and* $\gamma \in \mathbb{R}$*, write* $\rho = \frac{1}{2} + i\gamma$ *and set* $\operatorname{residueHarmonic}(x,\gamma) = -x^\rho/\rho$*.*

Theorem 6.3 (Zeros located as poles; `zeta_zero_located_as_pole`, `L_zero_located_as_pole`). *Let* $\rho = \frac{1}{2} + i\gamma$ *be a zero of* $L(\cdot,\chi)$ *at which* $L'(\rho,\chi) \neq 0$ *(a simple zero). Then* ρ *is a simple pole of the explicit-formula kernel* $-(L'/L)(s)\,x^s/s$*, with*

$$\lim_{s\to\rho}(s-\rho)\left(-\tfrac{L'(s,\chi)}{L(s,\chi)}\cdot\tfrac{x^s}{s}\right) \;=\; \operatorname{residueHarmonic}(x,\gamma) \;=\; -\frac{x^\rho}{\rho}.$$

The same statement holds verbatim for ζ*.*

Proof sketch. On the punctured neighborhood of ρ in $\{s \neq 1\}$, L is analytic with $L(\rho) = 0$ and $L'(\rho) \neq 0$, so $-(L'/L)$ has a simple pole at ρ with residue 1; multiplying by the analytic factor x^s/s scales the residue to x^ρ/ρ and the leading sign gives $-x^\rho/\rho$. □

7 Projection and reality on the critical line

The carrier is three-dimensional; the readout projects it to a one-dimensional coordinate. We first record that coordinate and the change of scale it carries (Section 7.1), then the reality of the completed object that the readout returns (Section 7.2).

7.1 The readout coordinate and its scaling

The carrier is built in its native gauge: the unit is $\pi/3$, and a full angular turn is six such units, $6\cdot(\pi/3) = 2\pi$ (`spin_phasor_period6`, Theorem 2.12). The readout that sends a three-dimensional harmonic value $t \in \mathbb{R}$ down to a one-dimensional coordinate composes a Cayley map [9] with a rescaling to the standard unit interval (`HarmonicProjection`):

$$\operatorname{toCircleAngle}(t) = 2\arctan t \in (-\pi,\pi), \qquad \operatorname{toLine}(\theta) = \frac{\theta+\pi}{2\pi} \in (0,1),$$

and projection = toLine∘toCircleAngle. The second map carries the change of scale: it sends the full turn 2π to length 1, so the six $\pi/3$ buckets of the gauge become six equal sub-intervals of the readout. Under this rescaling the coordinate is rigid — the Cayley map is odd (`toCircleAngle_odd`), the normalization is affine hence midpoint-preserving (`toLine_midpoint`), and the principal-arc ends $\theta = \pm\pi$ map to the two ends of the interval (`toLine_neg_pi`, `toLine_pi`) — so the centre of the gauge is carried to the centre of the interval.

Theorem 7.1 (Coordinate centre; `HarmonicProjection.projection_midline`). projection(0) *is the midpoint of the readout interval: the centre of the six gauge buckets, the angle $\theta = 0$, maps to the average of the two endpoint images (`toLine_midpoint_of_endpoints`).*

The inverse conversion is the same scaling read backwards: a readout coordinate u is the angle $\theta = 2\pi u - \pi$, the gauge reading $6u - 3$ in $\pi/3$ units, and the interval's centre is the gauge centre $\theta = 0$. This is a property of the readout coordinate and its scaling. The value the readout returns on the completed object is real — it lands on the real axis (Section 7.2).

7.2 Reality on the line

The supporting analytic facts are the conjugation (Schwarz reflection [9]) and functional symmetries of Λ.

Theorem 7.2 (Conjugation symmetry; `HelixCollapse.completedRiemannZeta_conj`). *For all $s \in \mathbb{C}$, $\overline{\Lambda(\overline{s})} = \Lambda(s)$.*

Proof sketch. On $\operatorname{Re} s > 1$ this is the real Dirichlet coefficients of Λ ($\overline{\Lambda(\overline{s})} = \Lambda(s)$ termwise, using $\overline{\pi^{-s/2}} = \pi^{-\overline{s}/2}$ and $\overline{\Gamma(\overline{s}/2)} = \Gamma(s/2)$); the identity extends to $\mathbb{C}$ by analytic continuation of Λ_0 (the pole-removed completion) via the identity theorem. □

Theorem 7.3 (Reality on the line; `completedRiemannZeta_critical_line_im_zero`). *For every $t \in \mathbb{R}$, $\operatorname{Im} \Lambda(\frac{1}{2} + it) = 0$.*

Proof sketch. On the line $\overline{\frac{1}{2} + it} = 1 - (\frac{1}{2} + it)$, so Theorem 7.2 together with $\Lambda(1 - s) = \Lambda(s)$ (`completedRiemannZeta_one_sub`) gives $\overline{\Lambda(\frac{1}{2} + it)} = \Lambda(\frac{1}{2} + it)$, i.e. a real value. □

Theorem 7.4 (Faithful projection for ζ; `faithful_projection_zeta`). *For every $\gamma \in \mathbb{R}$,*

$$\operatorname{Im} \Lambda\big(\tfrac{1}{2} + i\gamma\big) = 0 \qquad \textit{and} \qquad F_\eta(\gamma) = 0 \iff \zeta\big(\tfrac{1}{2} + i\gamma\big) = 0,$$

where $F_\eta(\gamma) = (1 - 2^{1-s})\zeta(s)$ at $s = \frac{1}{2} + i\gamma$ (`EtaTrivial.Feta`).

Theorem 7.5 (Located crossings lie on the line; `crossing_is_zero_on_line`). *If the eta-regularized limit of Theorem 5.3 vanishes at ordinate y, then $\zeta(\frac{1}{2} + iy) = 0$ and $\operatorname{Re}(\frac{1}{2} + iy) = \frac{1}{2}$.*

8 Faithfulness for every Dirichlet L-function and Tate completion

A single carrier serves all χ; the character weight $\chi(n)$ on each phasor is the only dial.

Theorem 8.1 (One carrier, all characters; `faithful_all_L`). *For every Dirichlet character χ:*

(i) *for $\operatorname{Re} s > 1$, $\sum_{n<N} \chi(n)\, n^{-s} \to L(s, \chi)$; and*

(ii) *on the critical line $\operatorname{Re} s = \frac{1}{2}$, the eta-twisted closed carrier `etaTwistClosed`(χ, s) vanishes if and only if $L(s, \chi) = 0$ (`etaTwistClosed_eq_zero_iff_critical`).*

Definition 8.2 (Completed carrier; `Tate.completedCarrier`). *For a Dirichlet character χ, set $\Lambda^{\mathrm{c}}_\chi(y) = \Lambda(\chi, \frac{1}{2} + iy)$ using the archimedean Γ-completion $\Lambda(\chi, s) = (\text{gamma factor}) \cdot L(s, \chi)$.*

Theorem 8.3 (Completion and functional equation; `faithful_all_L_completed`, `completedCarrier_eq_zero_iff`, `completed_functional_equation`). *For every* primitive *χ modulo q:*

(i) *the completion is zero-free off the zeros of L on the line:* $\Lambda^{c}_{\chi}(y) = 0 \iff L(\tfrac{1}{2} + iy, \chi) = 0$ *(the* Γ*-factor never vanishes for* $\operatorname{Re} s > 0$*); and*

(ii) *Tate's functional equation [10] holds:* $\Lambda(\chi, 1-s) = q^{\,s-1/2}\, W(\chi)\, \Lambda(\chi^{-1}, s)$*, where* $W(\chi) =$ `rootNumber`(χ)*.*

9 The double-ended carrier and conjugacy

The carrier is chiral: the right strand winds with spin $e^{-iy\log n}$, and its mirror, the left strand, winds with $e^{+iy\log n}$.

Theorem 9.1 (Conjugate chiralities; `both_helices_conjugate`)**.** *For every* $N \in \mathbb{N}$ *and* $y \in \mathbb{R}$*,*

$$\sum_{n<N} (-1)^{\,n+1}\, n^{-(1/2-iy)} \;=\; \overline{\sum_{n<N} (-1)^{\,n+1}\, n^{-(1/2+iy)}}\,.$$

Proof sketch. Termwise: $\overline{(-1)^{n+1}} = (-1)^{n+1}$ and $\overline{n^{-(1/2+iy)}} = n^{-(1/2-iy)}$ for the positive real base n (so $\arg n \neq \pi$), by `Complex.cpow_conj` and $\overline{n} = n$. $\square$

Corollary 9.2 (Equal modulus, identical crossing heights)**.** *The two strands have equal modulus at every* N*, so their limits vanish at the same ordinates* y*.*

Theorem 9.3 (Completed sides agree; `both_sides_cross_together`)**.** *For every* $y \in \mathbb{R}$*,* $\Lambda(\tfrac{1}{2} - iy) = \Lambda(\tfrac{1}{2} + iy)$*.*

Proof sketch. $\tfrac{1}{2} - iy = 1 - (\tfrac{1}{2} + iy)$, so this is $\Lambda(1-s) = \Lambda(s)$ (`completedRiemannZeta_one_sub`) on the line. $\square$

10 The Frobenius similitude and the determinant identity

The map $n \mapsto pn$ is the Frobenius self-similarity of the carrier. By Theorem 3.2 it acts on the carrier point $\mathrm{helixPt}(n) = \sqrt{n}\,\mathrm{wind}(n)$ as

$$\mathrm{helixPt}(pn) \;=\; \sqrt{pn}\,\mathrm{wind}(pn) \;=\; \underbrace{\sqrt{p}}_{\text{radial}}\;\underbrace{\mathrm{wind}(p)}_{\text{rotation}}\;\mathrm{helixPt}(n),$$

a similitude: radial scaling $\sqrt{p}$ (the area-law factor of Theorem 2.8) times a rotation by the unit-modulus winding $\mathrm{wind}(p)$. Under the bridge (Theorem 3.4) the rotation eigenphase at ordinate y is the spin $z = \mathrm{spin}(y,n) = e^{-iy\log n} \in \mathbb{T}$. The two chiralities (Theorem 9.1) carry z and $\overline{z}$.

Theorem 10.1 (Conjugate-eigenstate determinant; `frobenius_conjugate_det_one`)**.** *For every* $y \in \mathbb{R}$ *and* $n \in \mathbb{N}$*, with* $z = \mathrm{spin}(y,n) = e^{-iy\log n}$*,*

$$\det\begin{pmatrix} z & 0 \\ 0 & \overline{z} \end{pmatrix} \;=\; z\,\overline{z} \;=\; |z|^2 \;=\; 1.$$

Proof. The 2×2 determinant is $z \cdot \overline{z} - 0 \cdot 0 = z\,\overline{z}$. Since $\overline{z} = \overline{e^{-iy\log n}} = e^{\overline{-iy\log n}} = e^{+iy\log n}$ and $-iy\log n + \overline{-iy\log n} = 0$, we have $z\,\overline{z} = e^{-iy\log n}\,e^{+iy\log n} = e^0 = 1$. $\square$

Remark 10.2. *The transverse block* $\mathrm{diag}(z, \overline{z})$ *assembled from the two chiral eigenphases at a conjugate crossing has determinant* 1*; the radial scaling* $\sqrt{p}$ *of the similitude is carried separately and is not part of this block.*

10.1 The self-adjoint eigenstate

The determinant identity above is stated for the bare eigenphases $z, \overline{z} \in \mathbb{T}$. We realize them as the values of a genuine eigenstate of a self-adjoint generator. This is the operator *design*: it is real because its inputs are real, and it stands independently of any zero.

Definition 10.3 (Spectral wave; `spectralWave`). *For* $\gamma \in \mathbb{R}$, $\psi_\gamma(t) = e^{i\gamma t}$ $(t \in \mathbb{R})$.

Theorem 10.4 (Unit-modulus eigenstate of $D = -i\, d/dt$; `spectralWave_eigen`, `spectralWave_norm`). *For all* $t \in \mathbb{R}$, $-i\,\psi'_\gamma(t) = \gamma\,\psi_\gamma(t)$ *and* $\|\psi_\gamma(t)\| = 1$. *Thus* ψ_γ *is a unit-modulus eigenstate of the generator* $D = -i\, d/dt$ *with the real eigenvalue* γ.

The reality is structural, not an accident of ψ_γ. Multiplication by the real height, $\mathrm{vonNeumannOp}(\gamma) = \gamma{\cdot}\mathrm{id}$ on the fiber $\mathbb{C}$, is symmetric with eigenvalue γ (`vonNeumannOp_isSymmetric`, `vonNeumannOp_hasEigenvalue`); on the finitely-supported sequence space the real-diagonal operator $\mathrm{diagOp}(d)$ is symmetric (`diagOp_symmetric`), with the basis vectors as explicit eigenvectors carrying the real eigenvalues d_i (`diagOp_eigenvector`), and *every* eigenvalue is real (`diagOp_spectrum_real`) by von Neumann reality — a symmetric operator on a complex inner-product space has real eigenvalues (`symmetric_eigenvalue_real`). These are collected in `unconditional_frobenius_eigenstate` and `frobenius_spectralWave_eigenstate`.

Proposition 10.5 (The chiralities are eigenstate values; `spin_eq_spectralWave`, `conj_spin_eq_spectralWave`). *For every* $y \in \mathbb{R}$ *and* $n \in \mathbb{N}$, $\mathrm{spin}(y, n) = \psi_y(-\log n)$ *and* $\overline{\mathrm{spin}(y, n)} = \psi_y(\log n)$.

Theorem 10.6 (Determinant on the eigenstate values; `frobenius_spectralWave_det_one`). *For every* $y \in \mathbb{R}$ *and* $n \in \mathbb{N}$, $\det \mathrm{diag}\big(\psi_y(-\log n),\, \psi_y(\log n)\big) = 1$.

Theorem 10.7 (Frobenius unimodularity of the produced eigenstate; `frobenius_eigenstate_det_one`). *Fix* $\gamma \in \mathbb{R}$, $n \in \mathbb{N}$, *and* $\psi : \mathbb{R} \to \mathbb{C}$, *and* assume *the vanishing* $\Rightarrow$ *eigenstate link* $\psi = \psi_\gamma$. *Then* ψ *is a unit-modulus eigenstate of* $D = -i\, d/dt$ *with eigenvalue* γ, *and* $\det \mathrm{diag}\big(\psi(-\log n),\, \psi(\log n)\big) = 1$.

Remark 10.8 (The open link). *We exhibit the eigenstate* ψ_γ *for every real* γ *unconditionally (Theorem 10.4). What is* not *proved is that a carrier vanishing at height* γ produces *this eigenstate; that hypothesis* ($\psi = \psi_\gamma$) *is the one input of Theorem 10.7, left open. We do not derive that the vanishings are the spectrum of a canonical operator.*

11 The de Branges evaluation

The reality on the critical line (Section 7.2) and the eigenstate above sit inside the Hermite–Biehler / de Branges spectral framework. We record the framework, its reality and discreteness consequences, a worked example, and the change of variables that places the on-line cancellations as real spectral points.

Definition 11.1 (Structure function and components; `Estar`, `Acomp`, `Bcomp`, `IsHB`). *For* $E : \mathbb{C} \to \mathbb{C}$, *the reflection is* $E^*(z) = \overline{E(\overline{z})}$, *the components are* $A = \frac{1}{2}(E + E^*)$ *and* $B = \frac{i}{2}(E - E^*)$, *and* E *is* Hermite–Biehler *(*$\mathrm{IsHB}\, E$*) when* $\|E^*(z)\| < \|E(z)\|$ *for every* z *with* $\mathrm{Im}\, z > 0$.

Proposition 11.2 (Gram positivity; `gram_quadratic_form_nonneg`). *For vectors* v_i *in an inner-product space and scalars* c_i, $\sum_{i,j} \overline{c_i}\, c_j\, \langle v_i, v_j \rangle = \big\langle \sum_i c_i v_i, \sum_j c_j v_j \big\rangle \geq 0$ *— the positive-definite reproducing-kernel fact behind* $H(E)$.

Theorem 11.3 (Positivity forces a real spectrum; `hb_no_zero_upper`, `Acomp_zero_im_eq_zero`, `Bcomp_zero_im_eq_zero`)**.** *If* IsHB E *then* E *has no zeros in the open upper half-plane, and every zero of* A *and of* B *is real.*

Theorem 11.4 (Discreteness; `Bcomp_zeros_discrete`)**.** *If, in addition,* E *is entire, then the zeros of* B *are isolated: a real, discrete spectrum.*

Proposition 11.5 (Paley–Wiener example; `paleyWiener_isHB`, `paleyWiener_Bcomp`)**.** $E(z) = e^{-iz}$ *is Hermite–Biehler, and its* B*-component is* sin*, whose spectrum is* $\{k\pi\}$*.*

Theorem 11.6 (Critical-line change of variables; `deBranges_var_im`)**.** *For* $\rho \in \mathbb{C}$*,* $\operatorname{Im}\bigl(-i(\rho-\frac{1}{2})\bigr) = \frac{1}{2} - \operatorname{Re}\rho$*; so* $z = -i(\rho - \frac{1}{2})$ *is real iff* $\operatorname{Re}\rho = \frac{1}{2}$*.*

Corollary 11.7 (On-line cancellations are real spectral points; `criticalLine_deBranges_real`, `zeroHarmonic_deBranges_real`)**.** *A carrier vanishing at height* γ *sits at* $\rho = \frac{1}{2} + i\gamma$*, whose de Branges variable* $z = -i(\rho - \frac{1}{2}) = \gamma$ *is real. The same-height cancellations we find are real de Branges spectral points, evaluable in the reality/discreteness framework of Theorems 11.3–11.4.*

Remark 11.8 (Scope)**.** *Theorem 11.3 is conditional on* IsHB*. For a structure function built from* Λ*,* IsHB *— equivalently, no zeros off the critical line (`hb_no_zero_upper`) — is the Riemann Hypothesis; we neither assume nor prove it. By construction we evaluate only the on-line cancellations, whose de Branges variable is real unconditionally (Corollary 11.7); we make no claim about off-line zeros.*

12 Formalization

Every definition and theorem above is formalized in Lean 4 [12] against Mathlib [11], in the package `RequestProject` (files `ClosedForm.lean`, `HelixLogFreeFTA.lean`, `HelixCollapseReality.lean`, `LFunctionPhasor.lean`, `GeometricProjectionHolds.lean`, `Faithfulness.lean`, `AreaLaw.lean`, `UnconditionalFrobenius.lean`, `DeBranges.lean`). Each named theorem compiles with no `sorry`; the kernel axiom footprint reported by `#print axioms` (equivalently `lean_verify`) is

$$\{\ \text{propext},\quad \text{Classical.choice},\quad \text{Quot.sound}\ \},$$

the three standard Mathlib axioms, with no construction-specific axiom. The correspondence between the statements here and the Lean declarations is given by the identifiers in parentheses throughout.

The capstone identities — the strip extension (Theorems 5.1, 5.2, 5.3), the conjugacy of the two chiralities (Theorem 9.1), and the determinant identity (Theorem 10.1) — are collected in `Faithfulness.lean`; the geometry (Section 2) and the log-free winding (Section 3) live in `ClosedForm.lean` and `HelixLogFreeFTA.lean`, with the emergent area-law radius (Theorem 2.8, Definition 2.7) and the scale-critical exponent (Theorem 2.10) in `AreaLaw.lean`. The self-adjoint eigenstate design (Section 10.1) lives in `UnconditionalFrobenius.lean` and the Hermite–Biehler / de Branges framework (Section 11) in `DeBranges.lean`, with their connection to the carrier — the eigenstate identification (`spin_eq_spectralWave`, `frobenius_eigenstate_det_one`) and the on-line de Branges evaluation (`criticalLine_deBranges_real`, `zeroHarmonic_deBranges_real`) — collected in `Faithfulness.lean`.

Remark 12.1 (Scope)**.** *This paper presents a geometric tool: the double-ended chiral helix carrier and its phasor fiber, on which the vanishing points coincide with the zeros of* ζ *and the tested*

Dirichlet L-functions, verified numerically for the leading zeros. Its mathematical content is exactly the statements proved above — the carrier, the area-law derivation of the critical exponent (Theorem 2.10), the accumulation to $L(s,\chi)$ on $\operatorname{Re} s > 0$, the explicit-formula reading of the zeros as poles, the reality of the completed object on the critical line, the conjugacy of the two chiralities, the determinant identity of Theorem 10.1, the self-adjoint eigenstate realization of the chiral eigenphases (Section 10.1), and the de Branges evaluation placing the on-line cancellations as real spectral points (Section 11). Two links are taken as hypotheses and left open: that a vanishing produces *the eigenstate (Remark 10.8), and that the structure function for Λ is Hermite–Biehler, i.e. that there are no off-line zeros (Remark 11.8) — the latter being the Riemann Hypothesis, which this tool neither assumes nor proves.*

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